

Fady John Fayek

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📍 Heliopolis - Cairo

📅 2004-02-21

🇪🇬 Egyptian

🌐 LinkedIn

🐙 Github

🔗 Portfolio

Education

09/2022 – 07/2027 **Software Engineering**
Cairo, Egypt *Ain Shams University*
GPA 3.15/4

09/2008 – 07/2022 **Graduate**
Cairo, Egypt *College de la Sainte Famille*

Technical Skills

Programming Languages

C++, C, Java, Python, JavaScript, MATLAB, MS-SQL

Embedded Systems

Embedded C, ARM Cortex-M (Tiva C), UART, I2C, GPIO, timers, ISRs.

Web Development (MERN)

React.js, Node.js, Express.js, MongoDB, JWT Auth, REST APIs.

Tools & Platforms

Git/GitHub, Jira, Docker, VS Code, Postman, MATLAB, Simulink, Xilinx Vivado, QuestaSim, PySpark.

Soft Skills

Agile Methodology, Technical Leadership, Cross-functional Collaboration, Problem Solving.

Projects

- 04/2026 **License plate recognition** 🔗
Deep learning & Computer vision project
Egyptian car plate license extraction by applying filters and object detection using computer vision then recognising characters and objects using deep learning.
- 04/2026 **Distributed Load Balancing System for AI Workloads** 🔗
Distributed Computing course
Developed a distributed computing system for Large Language Model (LLM) inference and RAG, implementing load balancing (Round Robin, Least Connections) and GPU task scheduling to optimize performance and resource utilization.
- 12/2025 **Aegis Protocol: Reliable UDP Transport** 🔗
Python, Socket Programming, UDP
Engineered a reliable application-layer protocol over UDP with Stop-and-Wait ARQ, optimized binary packet design under strict MTU constraints, robust error control, and a custom simulation environment to validate performance under packet loss and latency.
- 12/2025 **Online Marketplace (MERN Stack)** 🔗
Full-Stack Web Application
Architected a full-featured MERN e-commerce platform with role-based access, secure JWT authentication, RESTful APIs for product and order management, and a Super User dashboard for real-time monitoring and system control.

12/2025	Embedded Door Locker Security System  <i>Embedded C & Microcontrollers</i> Designed and implemented a dual-ECU security system on Tiva C microcontrollers with UART-based communication, hardware abstraction for peripherals, and integrated safety features including auto-lock, password retry, and alarm mechanisms.
11/2025	Autonomous Dino Run Agent  <i>Reinforcement Learning (AI)</i> Developed a Python Q-Learning agent for the Chrome Dino game with a custom PyGame environment, dynamic state representation, and an Epsilon-Greedy strategy for autonomous optimal gameplay.
08/2025	E-Learning & Career Platform (ASP.NET MVC)  <i>MCV internship Capstone Project</i> Developed a full-featured LMS with ASP.NET Core MVC and SQL Server, including role-based portals, instructor dashboards, career simulation, secure admin panel, and an interactive Python Flask chatbot for web-based queries.
05/2025	App Rating & Success Predictor  <i>Machine Learning & Deep Learning</i> Developed a hybrid Random Forest–Deep Learning ensemble with stacking regression to predict Google Play Store app ratings, leveraging advanced feature engineering, text processing, and large-scale data analysis for improved accuracy.
05/2025	DC Motor Position Control System <i>Control Engineering Course</i> Built a PID-controlled feedback system with Arduino for precise DC motor positioning, tuning parameters via MATLAB/Simulink to optimize response and minimize error.

Professional Experience

07/2025 – 08/2025 Cairo, Egypt	MCV (Manufacturing Commercial Vehicles) Internship <i>Software Development Intern</i> Collaborated on full-stack development of “LuIZ,” a course registration web app using ASP.NET Core MVC with CSHTML/Bootstrap frontend, integrated a real-time Python/Flask AI chatbot with Google Gemini, implemented internal APIs, and delivered the project in an agile team environment.
07/2024 London, United Kingdom	University Of East London <i>ACE Summer School Program</i> Gained hands-on experience with AWS and Azure for IoT and Smart City solutions, performed large-scale data processing with PySpark and Python visualizations, and collaborated internationally to design and present innovative urban analytics projects.

Certificates & Volunteering

Big Data and Data Visualization On Site By University of East London	Artificial Intelligence By National Telecommunication Institute
IBM Data Science  By Coursera	Introduction to Generative AI Learning Path  By Coursera
Embedded Systems Level 1 By Edges Academy	ASU Career Center Volunteer By University of Ain Shams

Languages

Arabic Native	English IELTS B2 Academic	French Delf B2
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